

Certified by the Ceiling: Saturated Benchmarks Supply the No-Loss Claims Behind VLA Acceleration

Nihal Gunukula
Purdue University
ngunukul@purdue.edu

Abstract—Papers that make vision-language-action (VLA) policies cheaper to run justify their speedups by showing no loss on LIBERO, where base policies typically succeed in more than 90% of episodes. We show that such no-loss verdicts are supplied by tasks both settings already solve, and so say little about the tasks on which policies fail. Re-testing the best-documented case, a single denoising step on released flow-matching checkpoints, one step is equivalent to ten within ± 5 points on an eight-task LIBERO-10 battery for $\pi_{0.5}$, yet inconclusive on the four tasks classified as informative before the experiment ($[-2.0, +7.1]$ points). The pattern is systematic: of 18 equivalence certificates among 79 paired comparisons, the four saturated tasks keep 17 and the four informative tasks keep 1 at equal episode counts, while the informative tasks keep 57 of 59 significant differences. Where failures are common, the verdicts diverge: one step costs π_0 7.9 points ($p=0.00025$) and $\pi_{0.5}$ 4.7 points under large camera-viewpoint shifts ($p=0.0008$). The problem is widespread. Among 86 VLA efficiency papers claiming parity on LIBERO, 76% compare against a base policy at or above 90% success and 15 report any uncertainty, per-task result, or paired analysis. We propose a three-part report for no-loss claims, show that a relative failure margin alone is lenient in the opposite regime, and give the paired-episode budget that certification on informative tasks requires.

I. INTRODUCTION

Making vision-language-action (VLA) policies cheaper at inference is now a research area of its own: fewer denoising steps [1], [2], caching [3], adaptive computation [4], token pruning, and quantization, applied to released policies such as π_0 , $\pi_{0.5}$, and SmolVLA [5], [6], [7], [8]. These papers share one form of evidence: the accelerated policy shows no loss against its base on LIBERO [9], where the base already succeeds in nearly every episode. Released $\pi_{0.5}$, for example, loses 0.2 points at a single denoising step [3].¹

The claim of this paper is that such no-loss verdicts are largely supplied by tasks both settings already solve. A pair of episodes that both succeed is evidence that the settings agree on that episode, but it tightens an absolute-difference interval for the pooled average while carrying no information about the episodes where a setting could fail. Non-inferiority trials know this failure mode as the leniency of absolute margins at low event rates [10], [11]; saturated robot benchmarks keep evaluation permanently in that regime, and what it does to robot-policy claims has not been measured.

¹Per-episode logs for every reported number, the audit protocols and extractions, and the harness: <https://github.com/nihalgunu/roborigor>

We measure it on the best-documented case, a single denoising step on released flow-matching checkpoints. The favorable result reproduces with a formal equivalence test, but only once near-saturated tasks are admitted (Fig. 1a). Across every paired comparison in our grids, saturated tasks keep nearly all equivalence certificates and informative tasks keep almost none, while significant differences survive on the informative tasks (Fig. 1b). Where failures are common the verdicts diverge by policy and condition (Fig. 1c). An audit of 86 VLA efficiency papers shows that the field’s no-loss claims sit in exactly this regime (Fig. 2).

Why it matters. Authors of efficiency methods, reviewers who assess them, and practitioners who deploy them currently read “no loss on LIBERO” as evidence that a speedup is free. It is evidence that the speedup is free on tasks the policy already solves. Our reporting rule tells each of them what to ask for instead.

Contributions. (1) We show and quantify that equivalence verdicts in VLA evaluation are supplied by saturated tasks, with a task classification fixed before the experiment, an equal-size control over all task subsets, and sensitivity to interval and margin (Sec. IV). (2) We show that the VLA efficiency literature rests on such verdicts: an audit of 86 papers claiming no loss on LIBERO (Sec. VI). (3) We show what happens where failures are common, on three released policies and two LIBERO-Plus shifts, and derive a three-part report and the paired-episode budget for certifying no-loss claims (Secs. IV, V).

II. RELATED WORK

Equivalence and non-inferiority testing. Equivalence is conventionally shown with two one-sided tests [12], and reporting standards for equivalence and non-inferiority trials require the margin to be justified [13]. A margin on the absolute risk difference becomes lenient when the reference event rate is low, which motivates relative margins and non-inferiority frontiers [10], [11]; paired designs call for intervals that respect the pairing [14], [15]. We carry these results into robot-policy evaluation, where the event is failure and benchmark saturation drives its rate to a few percent.

VLA acceleration. Single- and few-step inference for flow VLAs has been obtained by distillation [1], one-step training [2], caching [3], and adaptive step counts [4]; regression

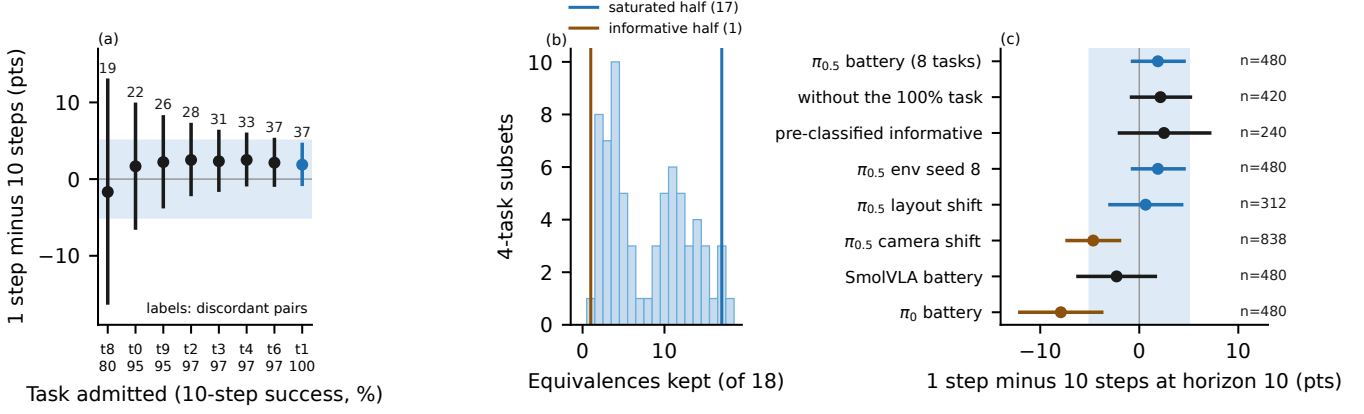


Fig. 1. Where equivalence certificates come from. All panels: one denoising step minus ten at execution horizon 10 (SmolVLA: minus its served default; paired Newcombe 95% intervals; shaded band: ± 5 -point equivalence margin). (a) $\pi_{0.5}$ on the LIBERO-10 battery, admitting tasks from least to most successful under ten steps (tick labels: task, ten-step success); labels count discordant pairs. The interval enters the margin (blue) only after the near-saturated tasks are admitted; the last, perfectly solved task adds 60 pairs and no discordant one. (b) The 18 comparisons in our grids that certify equivalence on the battery, re-tested on each of the 70 four-task subsets (240 pairs each): the saturated half keeps 17, the informative half 1. (c) The same contrast where failures are common. Brown: significant difference ($p < 0.05$); otherwise blue: equivalent at ± 5 ; black: inconclusive.

can match generative control [16], and execution horizon trades speed against reactivity [17], [18]. ActionCache reports $\pi_{0.5}$ collapsing at one step on VLABench [3], a far higher failure regime than LIBERO. We do not propose an acceleration method; we test what the evidence these methods report can establish.

Evaluation methodology. Seed variation can flip deep-RL conclusions [19], [20], and unreported variance inflates claimed progress in language models [21], [22]. In robotics, paired evaluation with posterior uncertainty is practiced at scale by the LBM study [23], sequential tests stop comparisons early [24], [25], best practice has been set out [26], and PhAIL audits real-robot VLA reporting [27]. Jiang et al. [28] show most LIBERO state-of-the-art claims cannot be significant given headline scores; vla-eval [29] shows undocumented evaluation details move LIBERO scores by tens of points; and LIBERO-Plus [30] and LIBERO-PRO [31] show large, uneven drops under perturbation. Those works concern claims that one policy is better than another; the no-loss claims on which efficiency results rest have not been examined.

III. PROTOCOL

Policies and battery. Released $\pi_{0.5}$ and π_0 LIBERO checkpoints (openpi) and SmolVLA, served through one websocket harness. The battery is eight LIBERO-10 tasks with 20 indexed initial states and 3 flow-sampling seeds ($n=480$ per setting). A design document written before any grid episode was run classified four battery tasks as informative (development-scan success in $[0.5, 0.95]$: tasks 0, 2, 8, 9) and four as saturated, for a variance study; using that classification as the equivalence subset here is our choice after the fact. Shifted conditions are fixed LIBERO-Plus variant sets: all 312 object-layout variants of libero_10 (one draw) and all 419 camera-viewpoint variants (two draws each). The study comprises 31,858 released episode records. **The contrast.** “One step” is a

single Euler step of the released flow head, without distillation or retraining, against ten steps, both at execution horizon $h=10$; SmolVLA’s reference arm is its served default, whose step count the records do not capture. **Statistics.** Comparisons are paired episode-for-episode on (initial state, sampling seed) or (variant, draw) and tested with the exact McNemar test [32]. Difference intervals are Newcombe’s paired score intervals [14]; unpaired Newcombe [33], paired Wald, and init-cluster bootstrap intervals are sensitivity analyses, and we state where verdicts differ. A comparison is a significant difference when $p < 0.05$; otherwise it is equivalent if its 95% interval lies inside ± 5 points (stricter than the 90% interval of two one-sided tests), and inconclusive if not. We report ± 3 , ± 7 , and a 90% rule as sensitivity. Counts of significant differences are unadjusted unless stated. Every number in this paper is regenerated from per-episode records by released scripts, and a consistency check pins the text to them.

IV. SATURATED TASKS SUPPLY THE CERTIFICATES

The favorable result reproduces. On the battery, $\pi_{0.5}$ succeeds in 0.965 of episodes at one step and 0.946 at ten (23–14 discordant pairs, $p=0.19$), and the difference interval $[-0.7, +4.5]$ points lies inside ± 5 . It does so again at a second environment seed ($[-0.7, +4.5]$), a genuinely different run: trajectory lengths differ across the seeds in 59% of paired episodes against 38% of fully fixed replicate cells, and the identical discordant totals are a coincidence (per task they differ, e.g. 3––1 vs. 7––4 on task 9). The step dial alone cuts measured inference latency by $1.4\times$ (66 vs. 94 ms per chunk on one A100). Taken at face value, this is the free speedup prior work reports [3], [1].

The certificate depends on near-saturated tasks. Under ten steps, six battery tasks succeed in 95–97% of episodes, one in 100%, and one (task 8) in 80%. Admitting tasks from the hardest to the easiest (Fig. 1a), the interval fits inside the

margin only at the end: with seven tasks it is $[-0.8, +5.2]$, and the eighth, perfectly solved task adds 60 pairs, no discordant ones, and the certificate. Under a 90% rule the certificate appears one near-saturated task earlier, so the point is not the single perfect task but the saturated tail. On the four informative tasks ($n=240$) the interval is $[-2.0, +7.1]$; on task 8 alone it is $[-16.1, +12.9]$ at $n=60$, with 9 discordant pairs won by one step and 10 by ten, so nothing here indicates a cost either: the question is simply not settled.

The pattern is systematic. The released grids contain 79 paired comparisons between dial settings measured on identical episodes ($\pi_{0.5}$ at two environment seeds and π_0). On the full battery 18 certify equivalence, all on $\pi_{0.5}$ and involving 13 distinct settings, so they are not independent, and 59 differ significantly (48 after Holm adjustment). Every four-task subset has the same 240 pairs (Fig. 1b). The four saturated tasks keep 17 of the 18 equivalences but only 38 of the 59 differences; the four informative tasks keep 1 equivalence and 57 differences; the median subset keeps 8 equivalences. The informative half is the unique minimum of the 70 subsets, and only 1 subset keeps more equivalences than the saturated half. Saturated tasks supply agreement; tasks where failures occur supply the evidence of difference. Among the 21 comparisons that change only the step count, 11 certify on the battery and 0 on the informative tasks, while 9 of their 10 significant differences survive there. The informative half’s result is carried largely by task 8, its one task with substantial headroom: subsets without it keep a median of 12 equivalences, subsets with it 4. The asymmetry holds under the unpaired Newcombe interval (14 on the battery; 14 saturated vs. 0 informative), the paired Wald interval (18; 17 vs. 3), a 90% rule (20; 18 vs. 5), and a ± 7 margin (20; 19 vs. 10); at ± 3 only 3 comparisons certify at all.

Why solved tasks tighten the interval. Take n pairs, of which b favor one setting and c the other. The estimated difference is $\hat{\delta} = (b - c)/n$ with standard error $\sqrt{b + c - (b - c)^2/n} / n$. Adding m pairs on which both settings succeed leaves b and c unchanged, so $\hat{\delta}$ and its standard error both shrink by the factor $n/(n + m)$ (the latter up to a correction that vanishes as n grows), whereas m further pairs at the same discordance rate would shrink the standard error only by $\sqrt{n/(n + m)}$. The McNemar statistic and the conditional odds ratio b/c are unchanged; what dilutes is the absolute margin. The same holds for the Newcombe intervals, whose Wilson components have half-width of order $\sqrt{f}/N$ for f failures among N episodes. The underlying reason is the base rate: a task succeeding in a fraction p of episodes has outcome variance $p(1-p)$, which vanishes at the ceiling. In our variance decomposition of the battery ($\pi_{0.5}$, up to 40 initial states $\times$ 10 seeds per task), task 8 carries total outcome variance 0.242 and every other task at most 0.063.

What certification requires. On the informative tasks the settings disagree on 11.7% of pairs. At that rate, certifying ± 5 points with 80% power when the true difference is zero takes about 520 independent paired episodes under our rule (Monte Carlo), against the 240 the battery allots; on task 8 (31.7%

TABLE I
THE THREE-PART REPORT FOR EVERY ONE-STEP VS. TEN-STEP CONTRAST ($h=10$, PAIRED: SMOLVLA AGAINST ITS SERVED DEFAULT). $b-c$: PAIRS WON BY ONE STEP VS. BY TEN. DIFFERENCE: PAIRED NEWCOMBE 95% INTERVAL. ODDS RATIO: EXACT CONDITIONAL INTERVAL. FAILURE RATIO: ONE-STEP OVER TEN-STEP FAILURES, INIT-CLUSTER BOOTSTRAP 95% INTERVAL.

Setting	n	$b-c$	Diff. (pts)	Odds ratio b/c	Failure ratio
$\pi_{0.5}$, battery	480	23-14	+1.9 [-0.7, +4.5]	1.64 [0.81, 3.45]	0.65 [0.31, 1.21]
without 100% task	420	23-14	+2.1 [-0.8, +5.2]	1.64 [0.81, 3.45]	–
informative tasks	240	17-11	+2.5 [-2.0, +7.1]	1.54 [0.68, 3.65]	0.70 [0.30, 1.30]
$\pi_{0.5}$, env. seed 8	480	23-14	+1.9 [-0.7, +4.5]	1.64 [0.81, 3.45]	0.67 [0.34, 1.11]
$\pi_{0.5}$, layout shift	312	16-14	+0.6 [-3.0, +4.3]	1.14 [0.52, 2.53]	0.93 [0.60, 1.39]
$\pi_{0.5}$, camera shift	838	45-84	-4.7 [-7.3, -2.0]	0.54 [0.36, 0.78]	1.08 [1.03, 1.14]
SmolVLA, battery	480	40-51	-2.3 [-6.2, +1.6]	0.78 [0.51, 1.21]	1.06 [0.95, 1.20]
π_0 , battery	480	33-71	-7.9 [-12.1, -3.8]	0.47 [0.30, 0.71]	1.53 [1.19, 1.96]

discordance) it takes about 1,280 against 60. Clustering by initial state would raise both.

No single margin scale suffices. The failure-rate ratio (failures under one step over failures under ten) ignores pairs both settings solve, so it gives the same answer on the battery (0.65 [0.31, 1.21]) and on the informative tasks (0.70 [0.30, 1.30]; init-cluster bootstrap): at an illustrative non-inferiority bound of 1.5 both certify, without dilution. It fails in the opposite regime. Under camera shift, where most episodes fail, the ratio is 1.08 [1.03, 1.14]: non-inferior at 1.5 although failures rise significantly. For π_0 it is 1.53 [1.19, 1.96] and does not certify. Absolute margins are lenient when failures are rare; relative margins are lenient when they are common.

A report for no-loss claims. We recommend that a no-loss claim report (i) the discordant pairs with the conditional odds-ratio interval, which solved tasks cannot move; (ii) the absolute interval on tasks classified as informative before the experiment; and (iii) a failure-rate-ratio interval against a declared bound. Table I applies all three to every contrast in this paper. The report costs nothing to compute from paired episodes that authors already run.

V. WHERE FAILURES ARE COMMON

Across policies. On released π_0 (same battery, $n=480$), one step costs 7.9 points: 0.773 vs. 0.852, 33-71 discordant pairs, $p=0.00025$, interval $[-12.1, -3.8]$. The full battery detects this cost: the ceiling does not hide large effects, it certifies agreement that informative tasks cannot confirm. On SmolVLA the contrast is inconclusive (0.625 vs. 0.648, $[-6.2, +1.6]$, $p=0.29$). Whether one step suffices therefore varies across policies; a few-shot fine-tuned π_0 evaluated elsewhere loses 1.4 points at one step on LIBERO-Spatial [34].

Under shift. On all 312 object-layout variants one step is equivalent for $\pi_{0.5}$ (0.920 vs. 0.913, $[-3.0, +4.3]$; not equivalent under the unpaired interval). Camera-viewpoint shift is LIBERO-Plus’s own robustness axis, and there both settings fail on more than half of episodes. A dedicated sample of all 419 camera variants, with two draws each (838 pairs), measures 0.4057 vs. 0.4523: a 4.7-point deficit, 45-84 discordant pairs, exact McNemar $p=0.0008$, paired interval $[-7.3, -2.0]$ (the conservative unpaired interval, $[-9.4, +0.1]$, includes zero). The cost sits in the largest viewpoint shifts: at LIBERO-Plus levels 4 and 5, one step wins 31 discordant pairs

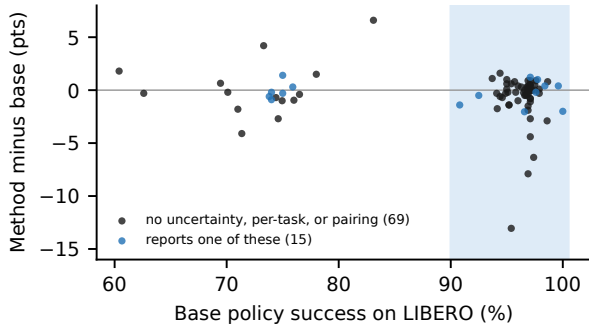


Fig. 2. No-loss claims in the 84 audited VLA efficiency papers with numeric LIBERO rates: base-policy success vs. method minus base. Shaded: base at or above 90%, where Sec. IV shows certificates are least informative. Blue: the claim reports uncertainty, per-task results, or a paired analysis.

and ten steps win 74, while at levels 1–3 the counts are 14 and 10. Prior work reports one step matching ten on LIBERO-Plus when pooled across perturbations for self-trained models [2]; on a released checkpoint, separated by axis and level, the cost is real.

VI. THE LITERATURE’S NO-LOSS CLAIMS

Following a written search and extraction protocol (one Semantic Scholar query, logged web and citation searches, and snowballing; 354 candidates screened in two stages), 86 VLA inference-efficiency papers claim parity or acceptable loss against their own base policy on LIBERO (Fig. 2). They span step reduction, distillation, caching, token pruning, quantization, early exit, and speculative decoding. The base succeeds in at least 90% of episodes in 76% of these claims (66% at 95%), and the median claimed gap is -0.2 points. Only 15 report any uncertainty, per-task result, or paired analysis for the claim, *one* reports a paired analysis, and 45% state no usable episode count. Of the 47 claims with counts, 38 would certify equivalence at ± 5 points from their own suite averages, and 14 of those cannot exclude 50% more failures (Katz interval on the failure-rate ratio). We also checked whether the same comparisons lose more on harder benchmarks the papers report; they do not (larger loss in 9 of 30 papers), though those benchmarks are chosen by the authors and are often small real-robot tests in which lower latency can itself raise success. The field’s no-loss claims thus sit where certificates are least informative; the audit does not show that the shortcuts they certify are costly. Extraction was LLM-assisted; direct text search of a seeded 20% sample confirmed 51 of 54 extracted fields, and manual checks the remaining 3.

Other reporting gaps. In a separate audit of 131 highlighted comparisons from 50 VLA papers, 34.4% state no usable rollout count and evaluation-seed counts appear for 31; among recomputable comparisons, 31.4% fail an exact test at the papers’ own rollout counts, concordant with Jiang et al. [28]. Of the 24 audited papers whose own method uses a flow or diffusion action head, 7 state the denoising step count used in evaluation, 6 the execution horizon, and at most 4 both,

so the two settings whose effects Sec. V measures are usually unrecoverable from the paper.

VII. LIMITATIONS

All results are simulation-only and LIBERO-family, on three flow-matching policies, two from one model family. The core demonstration rests on one policy: the equivalence certificates in our grids come from $\pi_{0.5}$ and share arms, the informative/saturated classification was made for a different purpose, and the informative half’s evidence rests mostly on one task. We test one acceleration, the single denoising step; the audit shows the other families make the same kind of claim but does not re-test them. The ratio bound of 1.5 is illustrative. Latency is inference time on one A100. The audits were single-rater against fixed rules and LLM-assisted over full text; the efficiency audit’s search could not use the arXiv API and its snowballing was limited to papers flagged during reading. Neither has yet been checked by a human second rater. All extractions ship with verbatim evidence.

VIII. CONCLUSION

A no-loss verdict on a saturated benchmark is valid for the average it certifies and says little about the tasks on which policies fail. Across our grids, saturated tasks keep nearly every equivalence certificate and informative tasks keep almost none, while differences survive where failures occur; where failures are common, one step costs π_0 7.9 points and $\pi_{0.5}$ 4.7 points under large camera shifts. The VLA efficiency literature makes its claims in exactly this regime. Authors should report discordant pairs, results on informative tasks, and a relative margin; reviewers should ask for them; and practitioners should treat “no loss on LIBERO” as a claim about tasks their policy already solves.

REFERENCES

- [1] W. Luan, J. Li, W. Zhao, W. Zhang, T. Wu, and R. Ma, “SnapFlow: One-step action generation for flow-matching VLAs via progressive self-distillation,” *arXiv:2604.05656*, 2026.
- [2] Y. Chen *et al.*, “Let it be simple: One-step action generation for vision-language-action models,” *arXiv:2606.05737*, 2026.
- [3] R. Oi, H. Otsuka, K. Matsushima *et al.*, “ActionCache: Training-free acceleration for vision-language-action models with action caching and refinement,” *arXiv:2607.06370v2*, 2026.
- [4] S. Han, Y. Han, and Y. Yi, “AdaVLA: Adaptive step flow matching for training-free acceleration of vision-language-action models,” *arXiv:2608.29208*, 2026.
- [5] K. Black *et al.*, “ π_0 : A vision-language-action flow model for general robot control,” *arXiv:2410.24164*, 2024.
- [6] Physical Intelligence, “ $\pi_{0.5}$: A vision-language-action model with open-world generalization,” *arXiv:2504.16054*, 2025.
- [7] M. Shukor *et al.*, “SmolVLA: A vision-language-action model for affordable and efficient robotics,” *arXiv:2506.01844*, 2025.
- [8] Physical Intelligence, “ $\pi_{0.7}$: A steerable model with emergent capabilities,” Whitepaper, <https://www.pi.website/blog/pi07>, 2026.
- [9] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks*, 2023.
- [10] M. Quartagno, A. S. Walker, A. G. Babiker, R. M. Turner, M. K. B. Parmar, A. Copas, and I. R. White, “Handling an uncertain control group event risk in non-inferiority trials: Non-inferiority frontiers and the power-stabilising transformation,” *Trials*, vol. 21, p. 145, 2020.

- [11] B. Redfors, “A case for using relative rather than absolute noninferiority margins in clinical trials,” *JACC: Advances*, vol. 3, no. 4, p. 100913, 2024.
- [12] D. J. Schuirmann, “A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability,” *Journal of Pharmacokinetics and Biopharmaceutics*, vol. 15, no. 6, pp. 657–680, 1987.
- [13] G. Piaggio, D. R. Elbourne, S. J. Pocock, S. J. W. Evans, and D. G. Altman, “Reporting of noninferiority and equivalence randomized trials: Extension of the CONSORT 2010 statement,” *JAMA*, vol. 308, no. 24, pp. 2594–2604, 2012.
- [14] R. G. Newcombe, “Improved confidence intervals for the difference between binomial proportions based on paired data,” *Statistics in Medicine*, vol. 17, no. 22, pp. 2635–2650, 1998.
- [15] T. Tango, “Equivalence test and confidence interval for the difference in proportions for the paired-sample design,” *Statistics in Medicine*, vol. 17, no. 8, pp. 891–908, 1998.
- [16] C. Pan *et al.*, “Much ado about noising: Dispelling the myths of generative robotic control,” *International Conference on Learning Representations (ICLR)*, 2026.
- [17] H. Wang *et al.*, “VLA knows its limits: Adaptive execution horizons for robot policies,” *arXiv:2602.21445*, 2026.
- [18] K. Black, M. Galliker, and S. Levine, “Real-time execution of action chunking flow policies,” *arXiv:2506.07339*, 2025.
- [19] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger, “Deep reinforcement learning that matters,” in *AAAI Conference on Artificial Intelligence*, 2018.
- [20] R. Agarwal, M. Schwarzer, P. S. Castro, A. C. Courville, and M. G. Bellemare, “Deep reinforcement learning at the edge of the statistical precipice,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [21] A. Hochlehnert, H. Bhatnagar, V. Udandara *et al.*, “A sober look at progress in language model reasoning: Pitfalls and paths to reproducibility,” *arXiv:2504.07086*, 2025.
- [22] E. Miller, “Adding error bars to evals: A statistical approach to language model evaluations,” *arXiv:2411.00640*, 2024.
- [23] TRI LBM Team, “A careful examination of large behavior models for multitask dexterous manipulation,” *Science Robotics*, 2026.
- [24] D. Snyder, A. J. Hancock, A. Badithela, E. Dixon, P. Miller, R. A. Ambrus, A. Majumdar, M. Itkina, and H. Nishimura, “Is your imitation learning policy better than mine? policy comparison with near-optimal stopping,” *arXiv:2503.10966*, 2025.
- [25] D. Snyder, A. Badithela, N. Matni, G. Pappas, A. Majumdar, M. Itkina, and H. Nishimura, “Beyond binary success: Sample-efficient and statistically rigorous robot policy comparison,” *arXiv:2603.13616*, 2026.
- [26] H. Kress-Gazit *et al.*, “Robot learning as an empirical science: Best practices for policy evaluation,” *arXiv:2409.09491*, 2024.
- [27] S. Arkhangelskiy, “PhAIL: A real-robot VLA benchmark and distributional methodology,” *arXiv:2605.29710*, 2026.
- [28] T. Jiang, X. Tan, S. Wheeler, L. Sun, T. W. Ayalew, and M. Walter, “What are we actually benchmarking in robot manipulation?” *arXiv:2606.04233*, 2026.
- [29] S. Choi, Y. Lee, Y. Park *et al.*, “vla-eval: A unified evaluation harness for vision-language-action models,” *arXiv:2603.13966v2*, 2026.
- [30] S. Fei *et al.*, “LIBERO-Plus: In-depth robustness analysis of vision-language-action models,” *arXiv:2510.13626*, 2025.
- [31] X. Zhou *et al.*, “LIBERO-PRO: Towards robust and fair evaluation of vision-language-action models,” *arXiv:2510.03827*, 2025.
- [32] Q. McNemar, “Note on the sampling error of the difference between correlated proportions or percentages,” *Psychometrika*, vol. 12, pp. 153–157, 1947.
- [33] R. G. Newcombe, “Interval estimation for the difference between independent proportions: Comparison of eleven methods,” *Statistics in Medicine*, vol. 17, pp. 873–890, 1998.
- [34] K. Chen, Z. Liu, T. Zhang *et al.*, “ $\pi_{\tau 1}$: Online RL fine-tuning for flow-based vision-language-action models,” *arXiv:2510.25889*, 2025.